# Virtualization

**Brett Johnson**

# What is virtualization?

# What is virtualization?

A process that allows a computer to share its hardware resources with multiple digitally separated environments.

Running a computer in a Computer

# What is virtualization?

- Provides an abstract layer between physical hardware (Host) and the virtual environment (VMs)

- Physical resources such as CPU, RAM, Storage, and Network usage are shared between multiple VMs

- Provides the ability to easily build and destroy different VMs without affecting the rest of the system

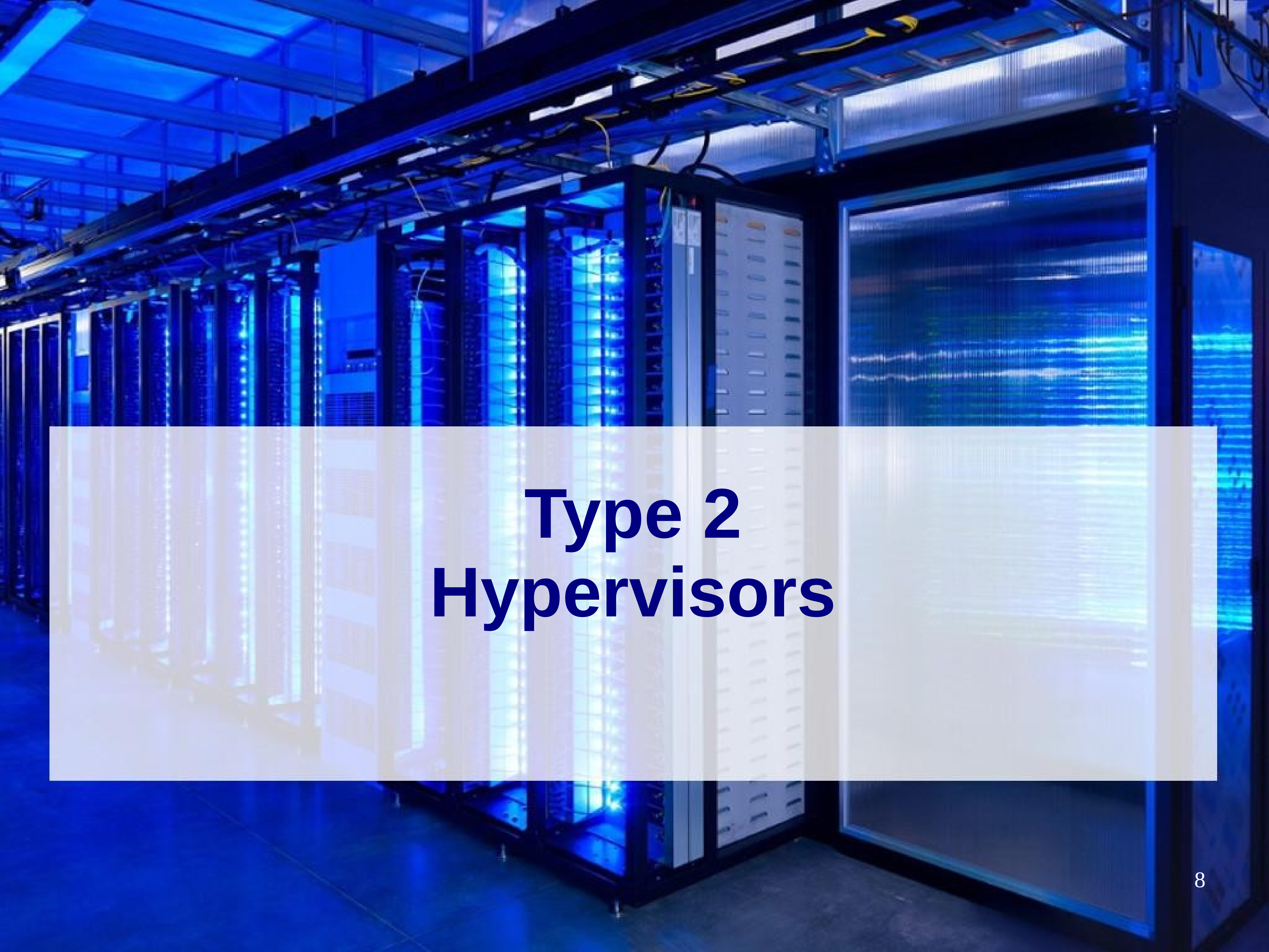
# Hypervisors

# Hypervisors

- Software that allows multiple VMs to run a single Host
- Manage the physical resources allocated to VMs
- Ensure that each VM is completely isolated

# Hypervisors

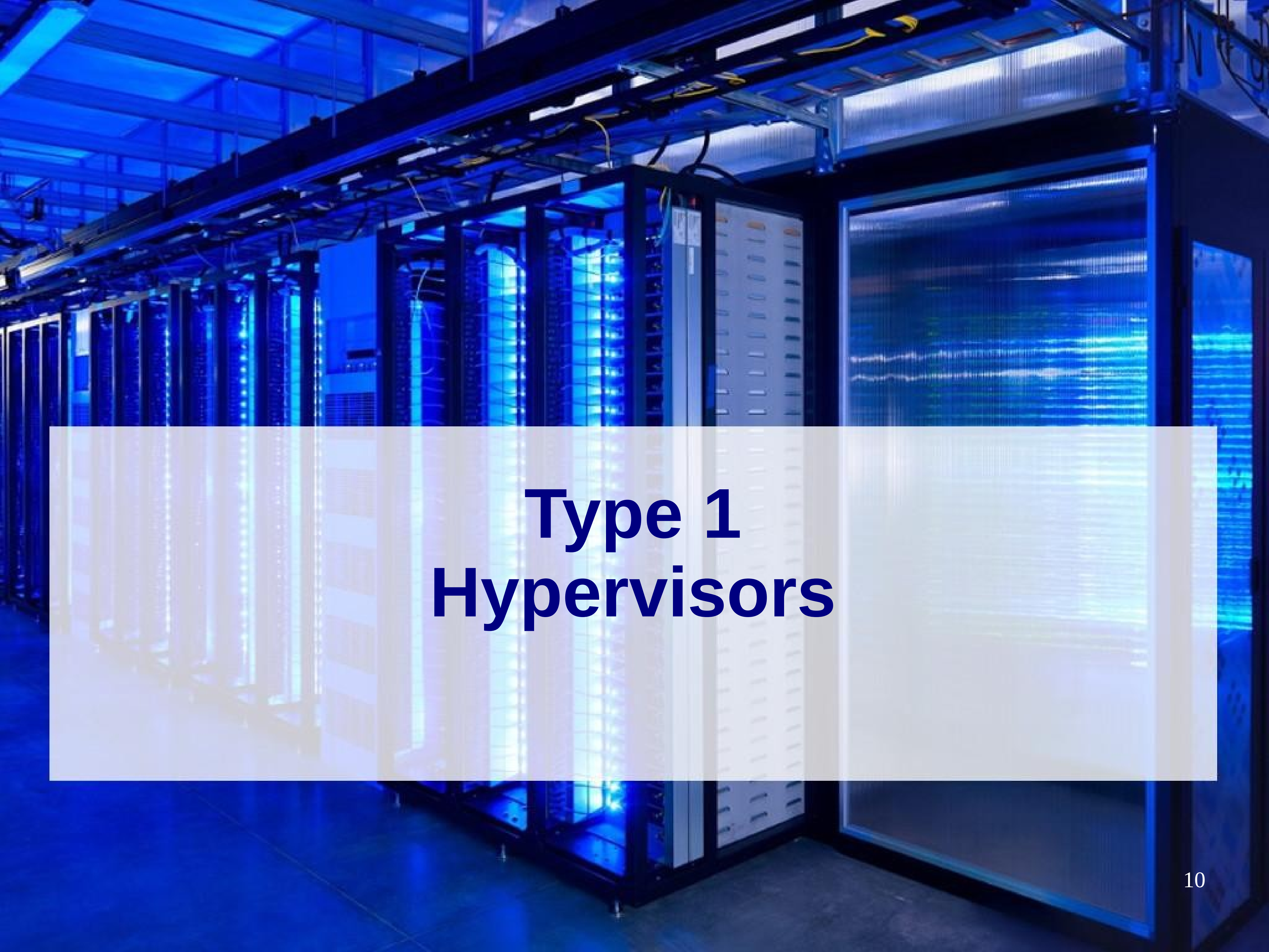
- Type 1 (Bare-metal or Native)
  - Runs directly on physical hardware
  - Is the Host Operating System (OS)
- Type 2 (Hosted)
  - Runs on top of a host OS



# Type 2 Hypervisors

# Type 2 Hypervisors

- Host is responsible for managing resources
- Great testing environments



# Type 1 Hypervisors

# Type 1 Hypervisors

- Is the OS
- Server applications
  - Data centers
  - High Performance
  - Run other Services

XCP-NG

PROXMOX

Microsoft
Hyper-V

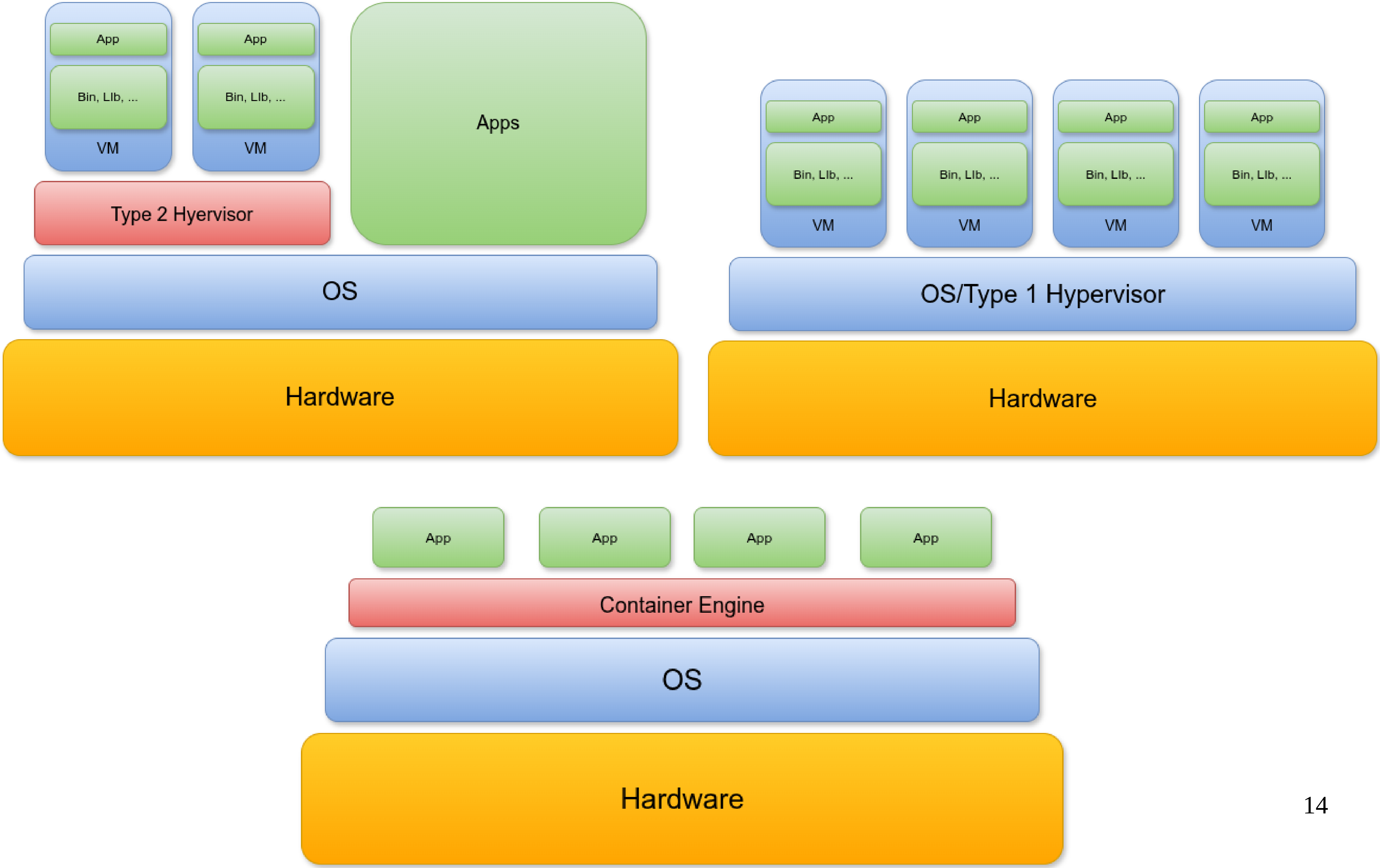
# Containerization

# Containerization

- A way of packaging software and its dependencies into a single, portable unit
    - Containers

- Lightweight
- Still isolated
- Very Portable
- Easy to Scale

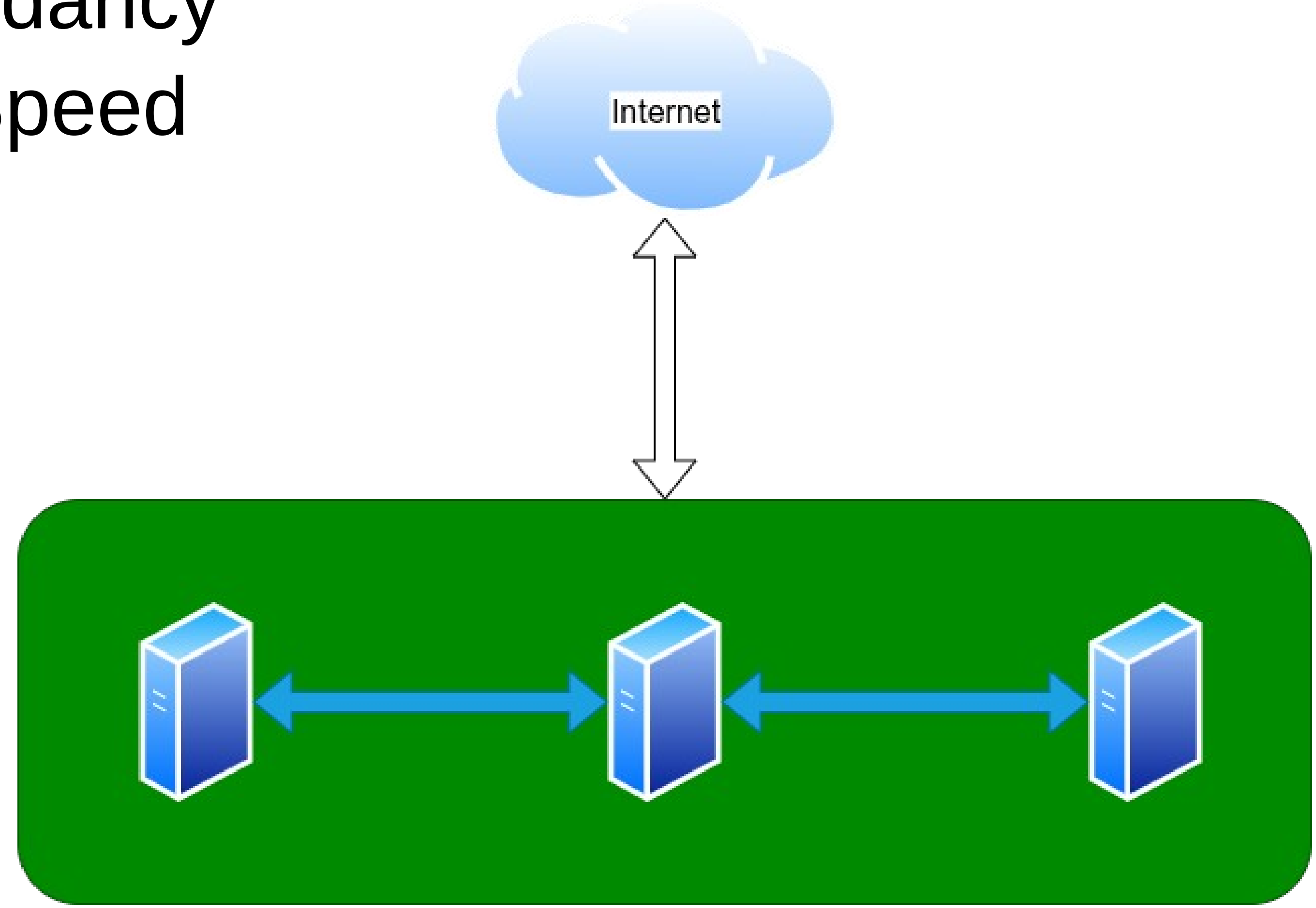
# Hypervisors



# Clusters & High Availability

# Clusters & HA

- Multiple hosts working together
- Redundancy
- High Speed



# Clusters & HA

- CEPH
  - Should use 3 or more hosts
  - Each host gives its ‘opinion’
  - Tie breaking vote wins

# Applications

# Applications

- Lab Environments (Build & Destroy)
- Sand-boxed Testing (Malware)
- Application Security (Flatpak)
- Services Ex: Nextcloud, PiHole, Website

# Getting Started

# VirtualBox

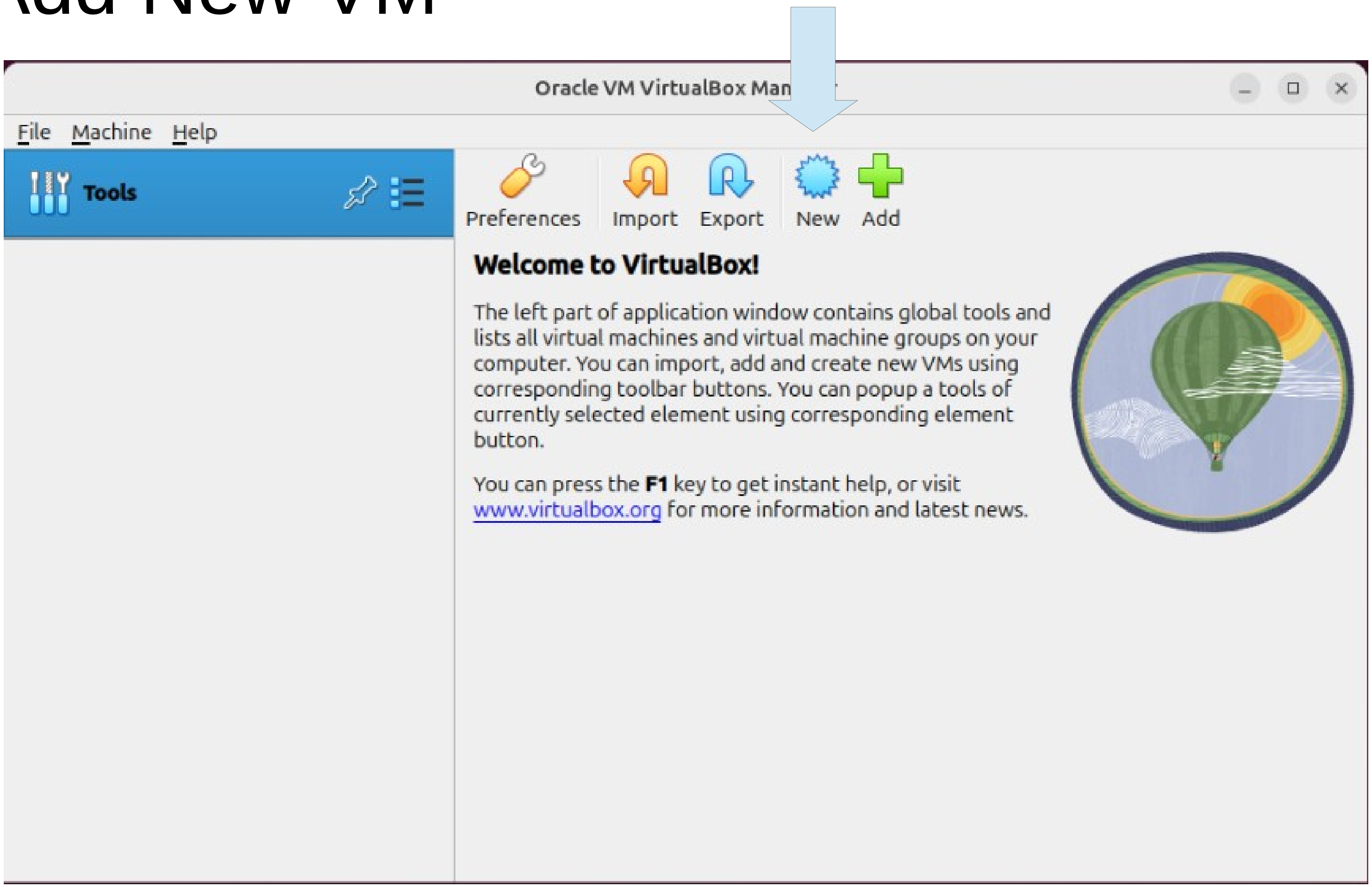
Installing VirtualBox
https://www.virtualbox.org/wiki/Downloads

(Windows)
1. Run through Startup Wizard
2. Restart Computer to finish driver installations

WARNING: Virtualization NEEDS to be turned ON in the BIOS

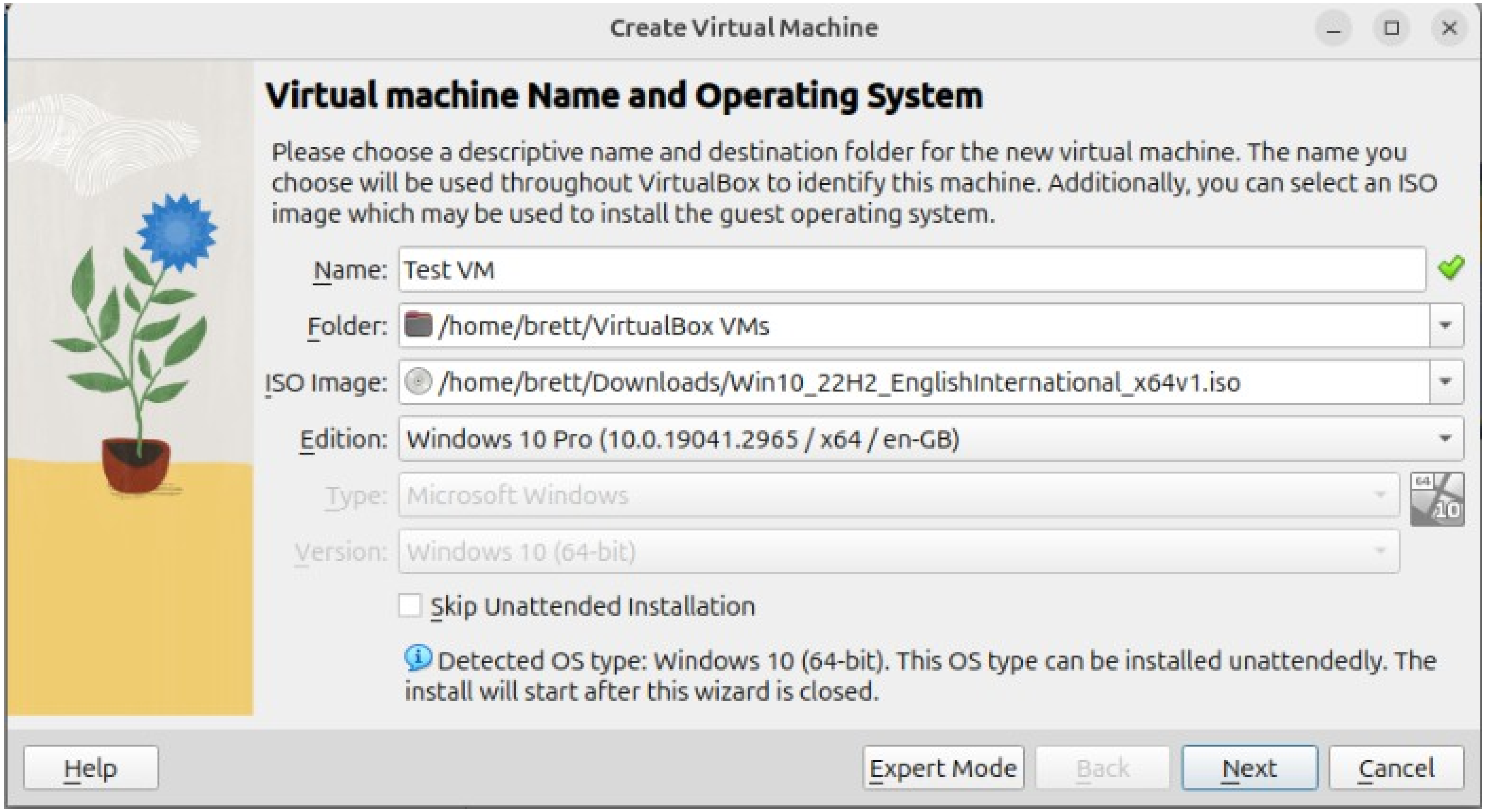
# VirtualBox

## Add New VM



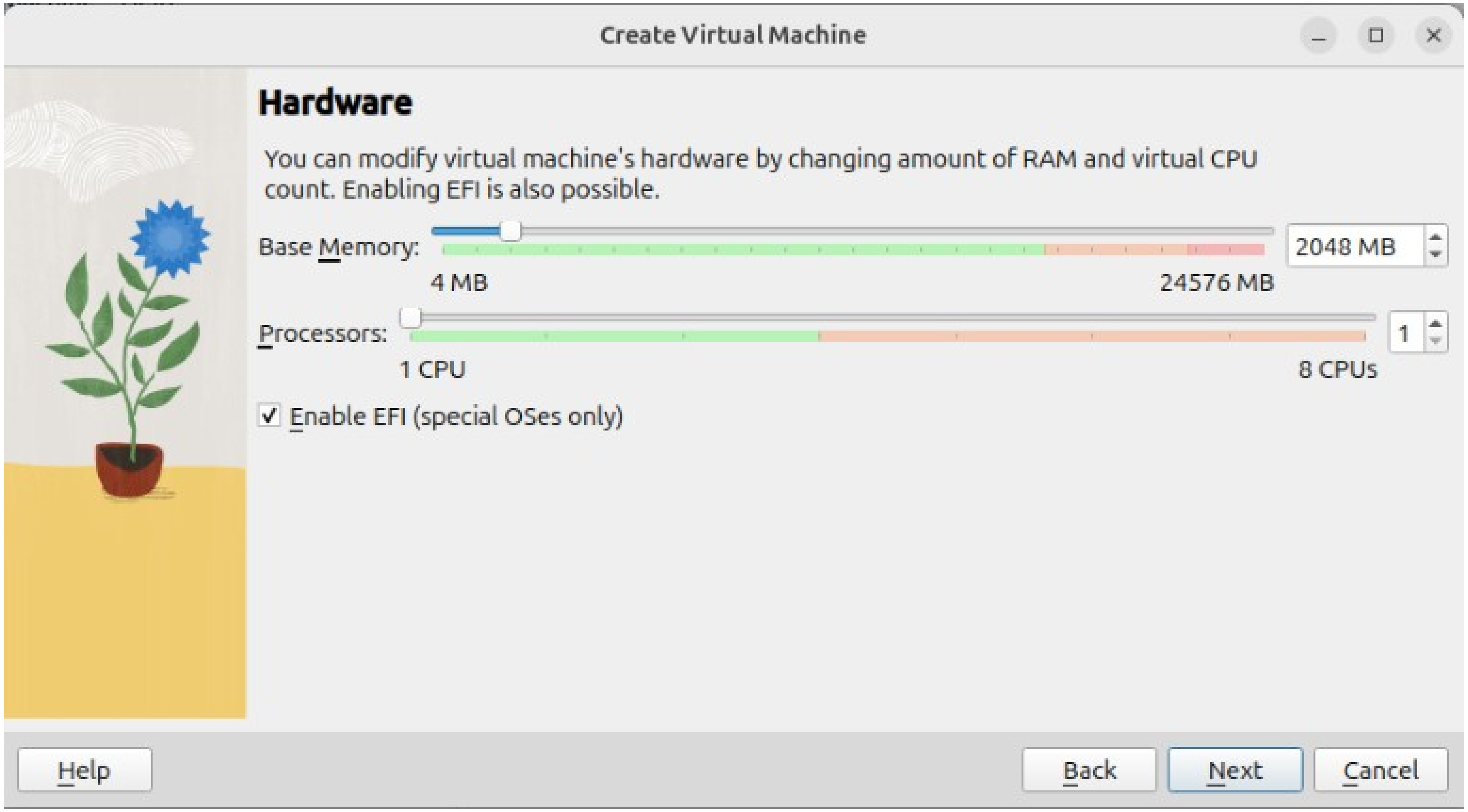
# VirtualBox

Fill out information
(leave “Skip Unattended Installation” unchecked for easy installation, NOTE: Product key is necessary if on windows)



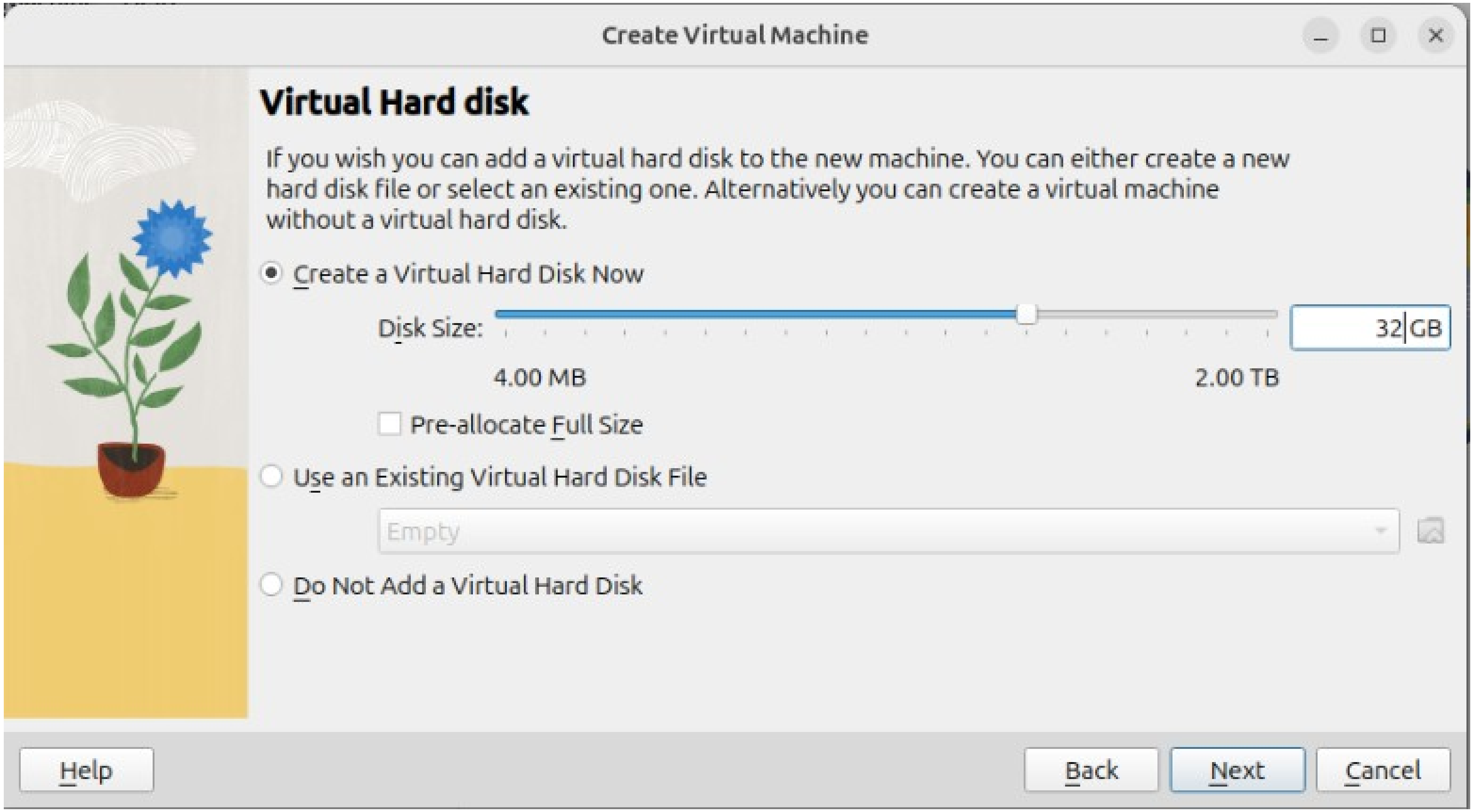
# VirtualBox

Set Memory Allocation (4 GiB – linux, 8 GiB – Windows)
Allocate vCPUs (2 – linux, 4 Windows)
Enable EFI



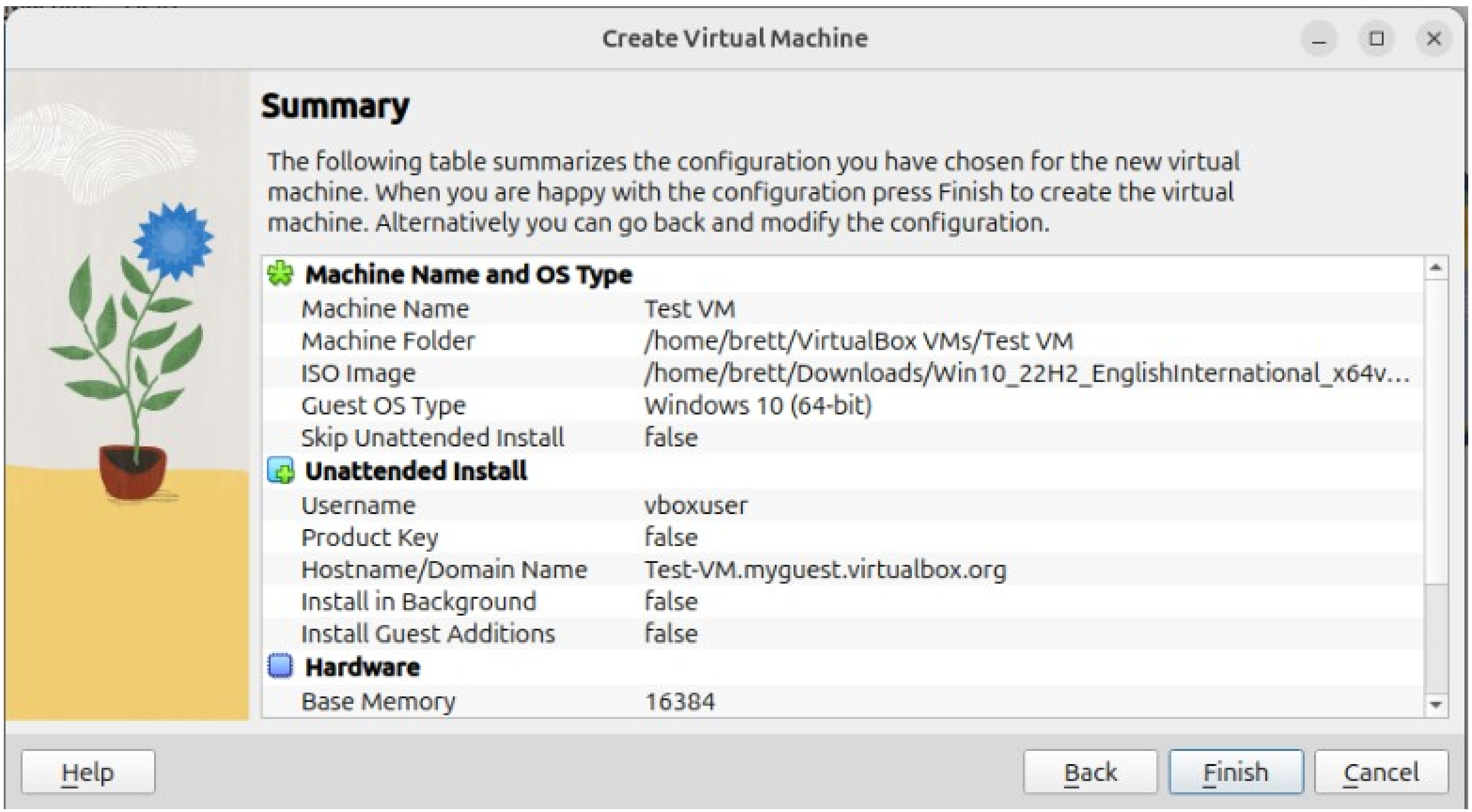
# VirtualBox

Create Disk (20 GiB – linux, 50 GiB – Windows)
LEAVE “Pre-allocate Full Disk Size” UNCHECKED



# VirtualBox

Double check everything and then click “Finish”



# Finishing Up

After running through installation

- Check for and run updates
- Install “Guest Drivers” (virtio)
- Check Display Settings

NOTE: GPU Pass-through is NOT possible with VirtualBox

# Questions?

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